Bridging Semantic and Kinematic Conditions with Diffusion-based Discrete Motion Tokenizer

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Research Questions

- How can we combine the strengths of continuous diffusion models (good at kinematic control) with discrete token-based generators (good at semantic conditioning) for motion generation?
- How can we design a motion tokenizer that uses fewer tokens while maintaining both semantic expressiveness and fine-grained motion reconstruction quality?
- How can we effectively handle both coarse semantic constraints and fine-grained kinematic constraints in a unified motion generation framework?

Core Idea

Existing motion generation methods face a fundamental trade-off: discrete token-based methods excel at semantic conditioning but struggle with fine-grained kinematic control, while diffusion models handle kinematics well but are less effective for semantic planning. The key insight is to create a division of labor where discrete tokens focus purely on semantic abstraction while delegating fine-grained reconstruction to a diffusion decoder. MoTok achieves this by using compact single-layer tokens that are decoded into per-frame conditioning signals for a diffusion model, rather than directly reconstructing motion. This allows the same discrete representation to work with both autoregressive and discrete diffusion planners while enabling precise kinematic control during the diffusion decoding stage.

Key Formula

Diffusion-based motion reconstruction

$$\hat{\mathbf{x}}_0 = f_\phi(\mathbf{x}_t, t, \mathbf{s}_{1:T})$$

The neural denoiser f_φ predicts clean motion $\hat{x}_0$ from noisy input x_t at diffusion timestep t, conditioned on per-frame signals $s_{1:T}$ decoded from discrete tokens. This is the core of MoTok's approach - using diffusion for fine-grained reconstruction while keeping tokens compact.

Fine-grained kinematic control refinement

$$\hat{\mathbf{x}}_k \leftarrow \hat{\mathbf{x}}_k - \eta \nabla_{\hat{\mathbf{x}}_k} \mathcal{L}_{\text{ctrl}}(\hat{\mathbf{x}}_k, \mathbf{c}^s_{1:T})$$

During diffusion denoising at step k, the motion estimate $\hat{x}_k$ is refined by gradient descent on control loss L_{ctrl} with respect to local conditions $c^s_{1:T}$. Parameter η controls refinement strength. This enables precise trajectory control without burdening the discrete planner.

Key Figure

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https://rheallyc.github.io/projects/motok

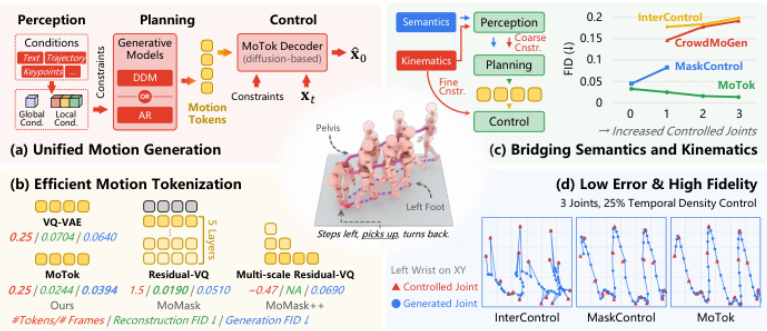

Fig. 1: **(a)** A unified Perception–Planning–Control pipeline for conditional motion generation. **(b)** MoTok enables compact motion tokenization with fewer tokens while maintaining competitive performance. **(c)** Bridging semantics and kinematics by applying coarse constraints in planning and fine constraints in motion control. MoTok maintains and improves fidelity as more joints are controlled, rather than compromising between controllability and realism. **(d)** Left-wrist XY trajectories with sparse control (red triangles). MoTok yields the most natural trajectory and best alignment.

Abstract. Prior motion generation largely follows two paradigms: continuous diffusion models that excel at kinematic control, and discrete token-based generators that are effective for semantic conditioning. To combine their strengths, we propose a three-stage framework comprising condition feature extraction *(Perception)*, discrete token generation *(Planning)*, and diffusion-based motion synthesis *(Control)*. Central to this framework is **MoTok**, a diffusion-based discrete motion tokenizer that decouples semantic abstraction from fine-grained reconstruction by delegating motion recovery to a diffusion decoder, enabling compact single-layer tokens while preserving motion fidelity. For kinematic conditions, coarse constraints guide token generation during *planning*, while fine-grained constraints are enforced during *control* through diffusion-based optimization. This design prevents kinematic details from disrupting semantic token planning. On HumanML3D, our method significantly improves controllability and fidelity over MaskControl while using only *one-sixth of the tokens*, reducing trajectory error from 0.72 cm to 0.08

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Figure 1: Shows the complete framework overview including the three-stage paradigm, token efficiency comparison, and the key finding that MoTok improves fidelity as more joints are controlled (unlike prior methods that degrade).

Limitations

- Method requires two-stage training (tokenizer then planner) which increases training complexity compared to end-to-end approaches
- Diffusion-based decoding still requires multiple denoising steps during inference, potentially slower than direct token decoding
- Framework is primarily evaluated on human motion - generalization to other types of sequential data is not demonstrated
- Relies on pre-defined condition taxonomy (global vs local) which may not capture all possible conditioning scenarios